



Figure 4: RAID synchronisation

access configuration, select RW access for FTP to the Internal PCs group. Now, you can access the FTP share with standard FTP clients from your internal network.

Accessing the FTP server from the Internet

To access this server from the Internet, you require to set up a small network with a firewall capable of port forwarding. IPCop is an excellent GNU firewall distribution for this purpose. Internal PCs/servers access the Internet via the firewall, which passes requests from internal PCs, to the requested Internet address, and passes replies back to the originating internal PCs. To allow external computers to access internal servers, we use a firewall feature called port forwarding. For more information, see http://en.wikipedia.org/wiki/Port_forwarding. Basically, connections to the firewall at certain port numbers used for different services can be passed on to servers behind the firewall, and the servers' responses relayed back to the computer on the Internet that requested the service. Some well-known ports are TCP 20, 21 for FTP data and FTP; TCP and UDP 53 for DNS; TCP 80 and 443 for HTTP/HTTPS; and UDP 161 for SNMP.

Prudent firewall installations first block all incoming Internet traffic to the internal network, and vice versa. Then, depending on the requirements, access is allowed for the requested services.

Configuring IPCop for port forwarding

Internet traffic to TCP ports 20 and 21 on the public Internet interface must be forwarded to the internal FTP (Openfiler) server.

Port forwarding configuration instructions will differ depending on firewall type. Figure 5 shows IPCop's port forwarding screen. Access can be restricted to a particular source IP address/range of IP addresses by entering the corresponding IP or network address in the Source IP field (see Figure 5). The ports are forwarded to the internal FTP server only if the originating IP address is the configured source IP.

That is all, folks! Now, the internal Openfiler FTP server can be accessed from pre-configured public IP addresses with any standard FTP client software such as Filezilla or CoreFTP.

Once installation is complete, don't forget to back up your system and LDAP configuration via *System-Backup/Restore – Download and Services-LDAP setup-Backup LDAP*.

For several important installation details, please refer to



Figure 5: IPCop port forward

the *LFY* August and September 2011 issues.

Real-life scenario: An IT-enabled Services (ITES) company daily received a lot of data from a client for processing. For this, the company had rented a huge space on an FTP server. The regular practice was that the client uploaded data on the FTP server, which the company downloaded to work on. After completing the work, the data was again uploaded on the FTP server for the client to download. Depending on data size, the upload and download took two to four hours on a 2 MBps leased line.

After installing an Openfiler FTP server with firewall port forwarding, the customer now uploads data directly to Openfiler, which is locally available to the company. Processed data goes back on the Openfiler server. This has saved the company rental costs of the FTP server, and most importantly, valuable upload/download time. 

References

- Installation screen-shots are available at <http://www.openfiler.com/products/screenshots>.
- Openfiler has a great community. The forum can be accessed at <http://openfiler.com/community/forums>.
- Download page: <http://openfiler.com/community/download/>
- The Openfiler administration guide is available for 40 Euros from <http://www.openfiler.com/docs/manual/>.
- IPCop Firewall: www.ipcop.org
- CoreFTP download page: <http://www.coreftp.com/download.html>
- Filezilla download page: <http://filezilla-project.org/download.php>
- Various protocols and port numbers: <http://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xml>

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